

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Scala for D.S Practical

Practical - 30

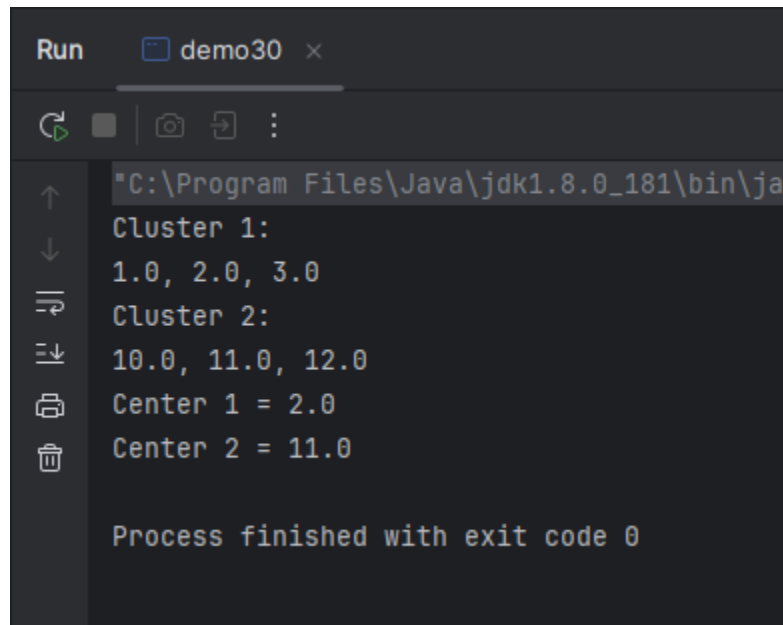
Roll No. : S025	Name: Abhishek Pal
Class : SY-Bsc.IT	Batch : 00
Date of Assignment : 29-09-2026	Date/Time of Submission : 29-09-2026

30) Cluster a dataset into two groups using k-means clustering in Breeze.

CODE :

```
build.sbt  demo30.scala x
1  import breeze.linalg._
2  object demo30 extends App{
3    val data = DenseVector(1.0, 2.0, 3.0, 10.0, 11.0, 12.0)
4
5    var center1 = 2.0
6    var center2 = 10.0
7
8    for (i <- 1 to 5) {
9      val cluster1 = data.toArray.filter(x =>
10        | math.abs(x - center1) < math.abs(x - center2)
11        | )
12
13      val cluster2 = data.toArray.filter(x =>
14        | math.abs(x - center2) <= math.abs(x - center1)
15        | )
16
17      center1 = cluster1.sum / cluster1.length
18      center2 = cluster2.sum / cluster2.length
19    }
20    println("Cluster 1:")
21    println(data.toArray.filter(x =>
22      | math.abs(x - center1) < math.abs(x - center2)
23      | ).mkString(", "))
24
25    println("Cluster 2:")
26    println(data.toArray.filter(x =>
27      | math.abs(x - center2) <= math.abs(x - center1)
28      | ).mkString(", "))
29
30    println("Center 1 = " + center1)
31    println("Center 2 = " + center2)
32  }
```

OUTPUT :



The screenshot shows a Java IDE's Run console. The title bar indicates the program is named 'demo30'. The console output is as follows:

```
*C:\Program Files\Java\jdk1.8.0_181\bin\ja  
Cluster 1:  
1.0, 2.0, 3.0  
Cluster 2:  
10.0, 11.0, 12.0  
Center 1 = 2.0  
Center 2 = 11.0  
  
Process finished with exit code 0
```